

The Monge Gap

Convexity, rearrangements, and kernel regression

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Abstract

The Monge gap of [Uscidda and Cuturi \(2023\)](#) compares the cost of a candidate map with the optimal transport cost between its input and output laws. Its nonnegativity is automatic, but its convexity is not. We characterize the smooth costs for which the gap is convex as a function of the map for every source measure: up to terms depending on one marginal only, the cost must be affine in the image variable. This class contains quadratic Mahalanobis costs and reverse Bregman divergences; among smooth displacement costs $h(x - y)$, it contains only quadratic functions, up to affine terms.

On the line, every convex displacement cost nevertheless has an explicit quantile formula. For a strictly convex cost its gap vanishes exactly on increasing maps, but the resulting monotonicity penalty is generally nonconvex. The quadratic case is exceptional: the gap is the deficit in the Hardy–Littlewood rearrangement inequality, or equivalently the support function of a translated permutahedron. This identifies its precise relation to isotonic regression and turns Monge-gap-regularized kernel regression into a finite convex program. For equally spaced inputs, the regularizer is a sum of pairwise hinge losses over all order inversions, yielding a standard quadratic program and a simple numerical experiment.

1 The exact Monge gap

The Monge gap separates marginal fitting from transport optimality. A fitting loss can require that a learned map reaches a prescribed target; the gap asks whether that map transports its own output law as cheaply as possible.

1.1 Definition and self-coupling representation

Let X be a standard Borel space, let $\mathcal{Y} \subset \mathbb{R}^m$, let $\rho \in \mathcal{P}(X)$, and let $T : X \rightarrow \mathcal{Y}$ be measurable. For a lower-semicontinuous cost $c : X \times \mathcal{Y} \rightarrow \mathbb{R} \cup \{+\infty\}$, write

$$\mathcal{L}_c(\alpha, \beta) := \inf_{\pi \in \Pi(\alpha, \beta)} \int c(x, y) \, d\pi(x, y)$$

whenever the relevant integrals are well defined.

Definition 1.1 (Monge gap [Uscidda and Cuturi, 2023](#)). The c -Monge gap of T relative to ρ is

$$\mathcal{M}_\rho^c(T) := \int c(x, T(x)) \, d\rho(x) - \mathcal{L}_c(\rho, T_\# \rho). \quad (1)$$

The first term is the cost of the graph coupling $\pi_T = (\text{Id}, T)_\# \rho$; the second is the smallest cost among all couplings with the same marginals. Consequently,

$$\mathcal{M}_\rho^c(T) \geq 0, \quad \mathcal{M}_\rho^c(T) = 0 \iff (\text{Id}, T)_\# \rho \text{ is optimal.} \quad (2)$$

The dependence of the second marginal on T obscures the geometry of the functional. Lifting every coupling back to two copies of the source removes this difficulty.

Lemma 1.2 (Lifting through a map). *For every $\pi \in \Pi(\rho, T_{\#}\rho)$, there exists $\eta \in \Pi(\rho, \rho)$ such that the map $(x, z) \mapsto (x, T(z))$ pushes η to π .*

Proof. Disintegrate ρ along T as $\rho(dz) = \int \rho_y(dz)(T_{\#}\rho)(dy)$, where ρ_y is concentrated on $T^{-1}(y)$, and disintegrate $\pi(dx, dy) = \pi_y(dx)(T_{\#}\rho)(dy)$. Then

$$\eta(dx, dz) := \int \pi_y(dx) \rho_y(dz) (T_{\#}\rho)(dy)$$

has both marginals equal to ρ , and $T(z) = y$ almost surely. $\square$

Proposition 1.3 (General self-coupling formula). *Assume that all terms below are finite. Then*

$$\mathcal{L}_c(\rho, T_{\#}\rho) = \inf_{\eta \in \Pi(\rho, \rho)} \int c(x, T(z)) d\eta(x, z), \quad (3)$$

$$\mathcal{M}_{\rho}^c(T) = \sup_{\eta \in \Pi(\rho, \rho)} \int [c(x, T(x)) - c(x, T(z))] d\eta(x, z). \quad (4)$$

Proof. Every self-coupling η induces an admissible coupling by $(x, z) \mapsto (x, T(z))$, while Lemma 1.2 shows that every admissible coupling has such a lift. This proves (3); subtracting it from the graph cost gives (4). $\square$

Formula (4) is always an optimality-gap representation. It is not generally a supremum of convex, let alone linear, functions of T . The next section identifies exactly when this cancellation does occur for smooth costs.

2 Which costs give a convex Monge gap?

Convexity for one fixed source may result from accidental symmetries. A useful cost-level statement should instead hold for every source measure. Under mild smoothness, this universal requirement has a sharp answer.

2.1 Affine differences characterize universal convexity

Call c an *affine-difference cost* if, for every x_0, x_1 , the map $y \mapsto c(x_0, y) - c(x_1, y)$ is affine. Equivalently, after choosing an origin in $\mathcal{Y}$, it can be written

$$c(x, y) = a(x) + b(y) - \langle A(x), y \rangle \quad (5)$$

for some scalar functions a, b and a map $A : \mathcal{X} \rightarrow \mathbb{R}^m$. The terms $a(x)$ and $b(y)$ depend on one marginal only and disappear from the gap.

Theorem 2.1 (Smooth convexity characterization). *Let $\mathcal{Y} \subset \mathbb{R}^m$ be open and convex, and assume that $c(x, \cdot) \in C^2(\mathcal{Y})$ for every x . The following statements are equivalent.*

- (i) *For every finitely supported source ρ , the map $T \mapsto \mathcal{M}_{\rho}^c(T)$ is convex on its finite-dimensional space of values.*
- (ii) *The same convexity holds for every equal-weight two-point source.*
- (iii) *Every difference $c(x_0, \cdot) - c(x_1, \cdot)$ is affine.*
- (iv) *The cost has the affine-difference form (5).*

For such a cost,

$$\mathcal{M}_\rho^c(T) = \sup_{\eta \in \Pi(\rho, \rho)} \int \langle A(x), T(z) - T(x) \rangle d\eta(x, z), \quad (6)$$

so the gap is convex. When $\mathcal{Y} = \mathbb{R}^m$ (or the admissible map class is stable under nonnegative scaling), it is also positively homogeneous in T .

Proof. Clearly (i) implies (ii). Fix x_0, x_1 and set $\rho = \frac{1}{2}(\delta_{x_0} + \delta_{x_1})$. A map is determined by $u = T(x_0)$ and $v = T(x_1)$. The two extreme self-couplings are the identity and the transposition, hence

$$\mathcal{M}_\rho^c(T) = \frac{1}{2} [g(u) - g(v)]_+, \quad g(y) := c(x_0, y) - c(x_1, y). \quad (7)$$

Suppose this function is convex in (u, v) . On the open set $g(u) > g(v)$, its Hessian is

$$\frac{1}{2} \begin{pmatrix} \nabla^2 g(u) & 0 \\ 0 & -\nabla^2 g(v) \end{pmatrix},$$

which must be positive semidefinite. Thus $\nabla^2 g(u) \geq 0$ at every point whose value exceeds another value, and $\nabla^2 g(v) \leq 0$ at every point whose value is below another one. A point that is neither a global minimum nor a global maximum satisfies both inequalities. At a global minimum or maximum, the missing inequality follows from the ordinary second-order extremum condition. Hence $\nabla^2 g = 0$ everywhere; g is affine. This proves (ii) $\Rightarrow$ (iii).

Choose a fixed $\bar{x}$. By (iii), $c(x, y) - c(\bar{x}, y) = a(x) - \langle A(x), y \rangle$ for suitable a, A ; taking $b(y) = c(\bar{x}, y)$ gives (iv). Finally, insert (5) into (4). Since both marginals of η equal ρ , the integrals of $b(T(x))$ and $b(T(z))$ cancel, which gives (6). It is a supremum of linear functionals of T , proving (i) and the final claim. $\square$

This theorem is a universal statement: a particular pair (c, ρ) may still produce a convex gap outside this class. Conversely, the two-point formula (7) is a cheap diagnostic that disproves universal convexity.

2.2 Examples and a counterexample

Quadratic and Mahalanobis costs. For $Q \geq 0$,

$$c_Q(x, y) = \frac{1}{2}(x - y)^\top Q(x - y)$$

has the form (5) with $A(x) = Qx$. The usual quadratic cost corresponds to $Q = \text{Id}$.

Reverse Bregman divergences. If F is differentiable and convex, then

$$D_F(y, x) := F(y) - F(x) - \langle \nabla F(x), y - x \rangle$$

is also an affine-difference cost, with $A(x) = \nabla F(x)$. Thus a genuinely nonquadratic cost can have a convex Monge gap. The orientation matters: $D_F(x, y)$ is generally nonlinear in the second argument in a way that does not cancel.

Corollary 2.2 (Smooth displacement costs). *Let $c(x, y) = h(x - y)$ with $h \in C^2(\mathbb{R}^d)$. The Monge gap is convex in T for every finitely supported source if and only if*

$$h(z) = \frac{1}{2}z^\top Qz + r^\top z + s \quad (8)$$

for a symmetric matrix Q . If h is convex, then necessarily $Q \geq 0$. The affine terms do not affect the gap.

Proof. Theorem 2.1 requires $\nabla_y^2 c(x, y) = \nabla^2 h(x - y)$ to be independent of x . Hence $\nabla^2 h$ is constant, which gives (8). The converse follows from the Mahalanobis example. $\square$

Example 2.3 (A convex cost with a nonconvex gap). Take $c(x, y) = (x - y)^4$ on the line and $\rho = \frac{1}{2}(\delta_{-1} + \delta_1)$. Let T_- , T_0 , and T_+ take the following values at $(-1, 1)$:

$$T_- = (-3/2, -2), \quad T_0 = (-1, -2), \quad T_+ = (-1/2, -2).$$

Then $T_0 = (T_- + T_+)/2$, while direct evaluation of the two possible assignments gives

$$\mathcal{M}_\rho^c(T_-) = 20.5, \quad qquad \mathcal{M}_\rho^c(T_0) = 32, \quad qquad \mathcal{M}_\rho^c(T_+) = 37.5.$$

Thus $32 > (20.5 + 37.5)/2 = 29$: even a smooth strictly convex displacement cost need not yield a convex Monge gap.

2.3 The quadratic support function

For $c_2(x, y) = \frac{1}{2} \|x - y\|^2$ and $\rho \in \mathcal{P}_2(\mathbb{R}^d)$, formula (6) becomes

$$\mathcal{M}_\rho^2(T) = \sup_{\eta \in \Pi(\rho, \rho)} \left\{ \int x \cdot T(z) \, d\eta(x, z) - \int x \cdot T(x) \, d\rho(x) \right\}. \quad (9)$$

It follows immediately that $\mathcal{M}_\rho^2$ is convex, nonnegative, subadditive, and positively homogeneous on $L^2(\rho; \mathbb{R}^d)$. Under Brenier's assumptions, its zero set has the expected geometric interpretation.

Proposition 2.4 (Zero set of the quadratic gap). *If ρ is absolutely continuous with respect to Lebesgue measure, then*

$$\mathcal{M}_\rho^2(T) = 0 \iff T = \nabla \varphi \quad \rho\text{-a.e. for some convex } \varphi.$$

For a finitely supported source, zero gap is equivalent to cyclical monotonicity of the pairs $(x_i, T(x_i))$, or to $T(x_i) \in \partial \varphi(x_i)$ for a convex φ .

Proof. By (2), zero gap means that the graph coupling is quadratically optimal. Brenier's theorem gives the first statement (Brenier, 1991; Villani, 2009); Rockafellar's theorem gives the finite cyclical-monotonicity statement (Rockafellar, 1966). $\square$

Convexity is in the free map values, not necessarily in a parametrization. If $T = T_\theta$ is a neural network, the composition $\theta \mapsto \mathcal{M}_\rho^2(T_\theta)$ is generally nonconvex.

3 One-dimensional rearrangement formulas

The line offers an explicit formula for every convex displacement cost. This is enough to recognize monotone maps, but not enough to make the penalty convex: Example 2.3 already occurs in one dimension.

3.1 General convex displacement costs

For a probability measure α on $\mathbb{R}$, denote its cumulative function by $F_\alpha(x) = \alpha((-\infty, x])$ and its generalized quantile by

$$Q_\alpha(s) := \inf\{x : F_\alpha(x) \geq s\}, \quad s \in (0, 1).$$

Proposition 3.1 (Closed one-dimensional Monge-gap formula). *Let $c(x, y) = h(x - y)$ with $h : \mathbb{R} \rightarrow \mathbb{R}$ convex, let $\rho \in \mathcal{P}(\mathbb{R})$, and set $\beta = T_{\#}\rho$. Under the corresponding integrability assumptions,*

$$\mathcal{M}_{\rho}^h(T) = \int h(x - T(x)) \, d\rho(x) - \int_0^1 h(Q_{\rho}(s) - Q_{\beta}(s)) \, ds. \quad (10)$$

If ρ is atomless, define the increasing rearrangement of T by

$$T^{\uparrow}(x) := Q_{\beta}(F_{\rho}(x)). \quad (11)$$

Then $(T^{\uparrow})_{\#}\rho = \beta$ and

$$\mathcal{M}_{\rho}^h(T) = \int [h(x - T(x)) - h(x - T^{\uparrow}(x))] \, d\rho(x). \quad (12)$$

If h is strictly convex, the gap vanishes exactly when T has a nondecreasing representative ρ -almost everywhere. For merely convex h , monotonicity is sufficient but need not be necessary.

Proof. For $x < x'$ and $y < y'$, convexity of h gives the Monge inequality

$$h(x - y) + h(x' - y') \leq h(x - y') + h(x' - y).$$

Indeed, the increment $r \mapsto h(r + y' - y) - h(r)$ is nondecreasing. Repeatedly uncrossing inverted pairs therefore shows that the quantile coupling $(Q_{\rho}, Q_{\beta})_{\#}\mathcal{U}(0, 1)$ is optimal. This proves (10). If ρ is atomless, $F_{\rho}(X)$ is uniform for $X \sim \rho$, so (11) realizes the quantile coupling as a map and gives (12). Strict convexity makes every genuine crossing strictly more expensive, yielding the final characterization. $\square$

For an equal-weight empirical source with $x_1 < \dots < x_n$, write $t_i = T(x_i)$ and let $t_{(1)} \leq \dots \leq t_{(n)}$ be their order statistics. Then

$$\mathcal{M}_{\rho}^h(t) = \frac{1}{n} \sum_{i=1}^n [h(x_i - t_i) - h(x_i - t_{(i)})]. \quad (13)$$

Thus the gap is evaluated by one sort, in $O(n \log n)$ time. This is an exact penalty for violations of increasing order when h is strictly convex, but Corollary 2.2 shows that it is generally nonconvex in t unless h is quadratic up to affine terms.

3.2 Quadratic cost, covariance deficit, and permutahedra

For $h(r) = r^2/2$, all target second moments cancel. Proposition 3.1 simplifies to

$$\mathcal{M}_{\rho}^2(T) = \int_0^1 Q_{\rho}(s)Q_{T_{\#}\rho}(s) \, ds - \int xT(x) \, d\rho(x). \quad (14)$$

The first term is the largest possible cross-moment, or covariance, among all couplings of ρ and $T_{\#}\rho$. Hence the one-dimensional quadratic Monge gap is precisely the deficit in the Hardy–Littlewood rearrangement inequality (Hardy et al., 1952). “Rearrangement deficit” or “comonotonicity deficit” describes this quantity accurately; there does not appear to be a separate canonical name for this exact functional.

For empirical data, let $x = (x_1, \dots, x_n)^{\top}$ and $t = (t_1, \dots, t_n)^{\top}$, with $x_1 < \dots < x_n$. Define the permutahedron

$$\mathfrak{P}(x) := \text{conv}\{P_{\sigma}x : \sigma \in \mathfrak{S}_n\}$$

and the support function $\sigma_C(t) := \sup_{q \in C} q^\top t$. The rearrangement inequality gives

$$\mathcal{M}_\rho^2(t) = \frac{1}{n}(x^\top t^\uparrow - x^\top t) = \frac{1}{n}(\sigma_{\mathbb{P}(x)}(t) - x^\top t) = \sigma_{D_x}(t), \quad D_x := \frac{\mathbb{P}(x) - x}{n}. \quad (15)$$

This proves convexity directly. If the entries of t are distinct, a subgradient is $(q^\star - x)/n$, where $q^\star$ permutes the entries of x into the rank order of t . Moreover,

$$\mathcal{M}_\rho^2(t) = 0 \iff t_1 \leq \dots \leq t_n. \quad (16)$$

There is an especially transparent formula on a uniform grid.

Proposition 3.2 (Pairwise inversion formula). *If $x_i = x_1 + (i - 1)\Delta$ with $\Delta > 0$, then*

$$\mathcal{M}_\rho^2(t) = \frac{\Delta}{n} \sum_{1 \leq i < j \leq n} (t_i - t_j)_+. \quad (17)$$

For $n = 2$, this reduces to $\mathcal{M}_\rho^2(t) = \frac{x_2 - x_1}{2}(t_1 - t_2)_+$.

Proof. Bubble-sorting t exchanges each inverted pair exactly once. Swapping two adjacent inverted values $u > v$ increases $x^\top t$ by $\Delta(u - v)$. Summing these increments over all inversions and dividing by n gives (17). $\square$

Formula (17) is the zero-margin pairwise ranking hinge summed over every inversion. It is therefore a polyhedral exact penalty for the monotone cone.

3.3 What is the relation with isotonic regression?

Classical least-squares isotonic regression computes the metric projection

$$\Pi_C(z) := \arg \min_{t_1 \leq \dots \leq t_n} \frac{1}{2} \|t - z\|^2, \quad C := \{t : t_1 \leq \dots \leq t_n\}. \quad (18)$$

The Monge gap and isotonic regression have the same target set C , but they are not the same operation:

- rearrangement sorts the existing values and preserves their empirical distribution;
- isotonic regression moves and pools values to minimize a metric fitting error;
- the Monge gap measures the improvement obtained by rearranging the graph coupling, rather than the distance from t to C .

The precise computational link goes through the permutahedron. Since $\mathcal{M}_\rho^2 = \sigma_{D_x}$, the proximal map satisfies

$$\text{prox}_{\tau \mathcal{M}_\rho^2}(z) = z - \tau \Pi_{D_x}(z/\tau) = z - \Pi_{\tau D_x}(z). \quad (19)$$

Projection onto a permutahedron reduces, after sorting, to isotonic optimization and can be solved by pool-adjacent-violators-type algorithms (Lim and Wright, 2016). Thus isotonic regression is an algorithmic primitive for proximal computations with the quadratic Monge gap, even though the gap itself is not an isotonic projection loss.

4 Kernel regression with a Monge-gap penalty

The preceding formulas make the proposed experiment tractable without any transport solver. Consider scalar regression data $x_1 < \dots < x_n$ and $y_1, \dots, y_n$, and a scalar RKHS $\mathcal{H}_k$. For the quadratic Monge gap relative to $\hat{\rho} = n^{-1} \sum_i \delta_{x_i}$, define

$$\min_{T \in \mathcal{H}_k} \left\{ \sum_{i=1}^n |T(x_i) - y_i|^2 + \lambda \|T\|_{\mathcal{H}_k}^2 + \gamma \mathcal{M}_{\hat{\rho}}^2(T) \right\}, \quad \lambda > 0, \quad \gamma \geq 0. \quad (20)$$

4.1 Convexity and the kernel reduction

All three terms in (20) are convex in T , and the RKHS term is strictly convex. Hence the problem has at most one minimizer; the usual coercivity and lower-semicontinuity assumptions give existence. More importantly, the kernel trick is exact.

Proposition 4.1 (Representer theorem and finite convex program). *Every minimizer of (20) has the form*

$$T(\cdot) = \sum_{j=1}^n a_j k(x_j, \cdot). \quad (21)$$

Let $K_{ij} = k(x_i, x_j)$, $a = (a_j)_j$, and $y = (y_i)_i$. The coefficients solve the finite convex problem

$$\min_{a \in \mathbb{R}^n} \left\{ \|Ka - y\|^2 + \lambda a^\top Ka + \frac{\gamma}{n} [\sigma_{\mathbb{P}(x)}(Ka) - x^\top Ka] \right\}. \quad (22)$$

If K is positive definite, this program has a unique coefficient vector.

Proof. Decompose any $T \in \mathcal{H}_k$ into its orthogonal projection onto $\text{span}\{k(x_i, \cdot)\}_{i=1}^n$ and an orthogonal remainder. The remainder vanishes at every x_i , so it changes neither the data term nor the Monge gap, while it strictly increases the squared RKHS norm unless it is zero. This proves (21), the classical representer argument (Kimeldorf and Wahba, 1971; Schölkopf et al., 2001). The identities $T(x_i) = (Ka)_i$ and $\|T\|_{\mathcal{H}_k}^2 = a^\top Ka$, together with (15), give (22). Convexity follows because a support function composed with a linear map is convex. $\square$

The support function is evaluated by sorting Ka , in $O(n \log n)$ time. A subgradient is obtained from the same ranks, so bundle, subgradient, and proximal methods require no $n \times n$ transport optimization.

Uniform-grid quadratic program. If the inputs are equally spaced with step Δ , introduce variables ξ_{ij} for $i < j$. Proposition 3.2 turns (22) into

$$\begin{aligned} \min_{a, \xi} \quad & \|Ka - y\|^2 + \lambda a^\top Ka + \frac{\gamma \Delta}{n} \sum_{i < j} \xi_{ij}, \\ \text{subject to} \quad & \xi_{ij} \geq (Ka)_i - (Ka)_j, \text{quad} \xi_{ij} \geq 0 \quad (i < j). \end{aligned} \quad (23)$$

This is a convex quadratic program with n kernel coefficients and $n(n-1)/2$ scalar slack variables. The quadratic number of slacks is suitable for a moderate experiment; sorting-oracle methods are preferable at large n . Since the penalty is polyhedral and vanishes exactly on the monotone cone, standard exact-penalty theory also implies that, under the usual constraint qualification, a finite sufficiently large γ recovers the hard monotone RKHS fit.

Dual projection viewpoint. If K is positive definite, use the fitted values $t = Ka$ and set

$$Q := I + \lambda K^{-1}, \quad D_x := \frac{\mathfrak{P}(x) - x}{n}.$$

For $\gamma > 0$, Fenchel duality gives

$$d^\star = \arg \min_{d \in D_x} \left(d - \frac{2y}{\gamma} \right)^\top Q^{-1} \left(d - \frac{2y}{\gamma} \right), \quad t^\star = Q^{-1} \left(y - \frac{\gamma}{2} d^\star \right). \quad (24)$$

Indeed, write $\sigma_{D_x}(t) = \max_{d \in D_x} d^\top t$ and minimize the resulting strongly convex quadratic in t . Equation (24) is a projection onto a translated permutahedron in the kernel-induced metric Q^{-1} . When that metric is diagonal, this is a separable isotonic optimization problem. For a general kernel, Q^{-1} is dense, so ordinary pool-adjacent violators is not by itself a complete solver; the quadratic program (23) or a general metric-projection method is still needed.

4.2 A numerical feasibility experiment

Figure 1 solves (23) for 55 equally spaced inputs, a Gaussian kernel, and several values of γ . The observations deliberately follow a smooth but nonmonotone trend. Increasing γ progressively removes order inversions; at the largest displayed value the solution agrees numerically with the hard monotone RKHS fit. The experiment is generated by `kernel_monge_regression.py` in this directory.

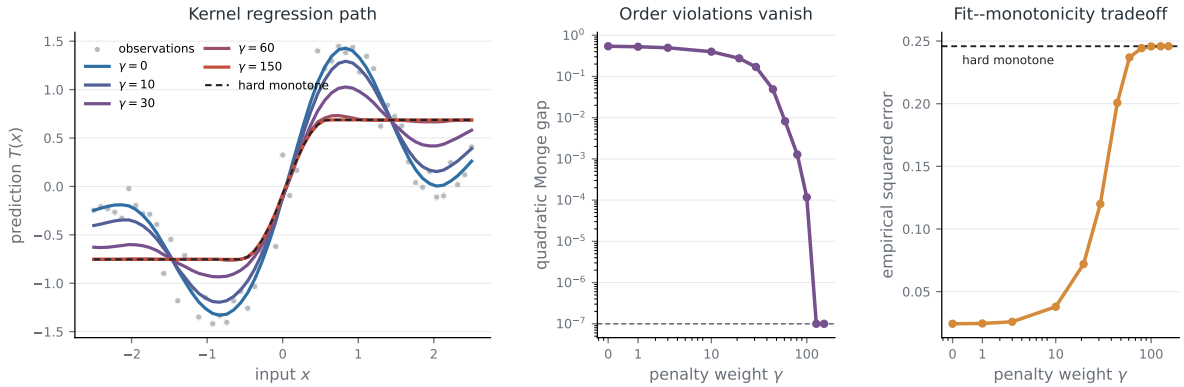


Figure 1: *Monge-gap-regularized kernel regression is a convex monotonicity path.* Left: Gaussian-RKHS fits evolve from an unconstrained smooth regressor (blue) to the hard monotone fit (red) as γ increases. Center: the exact empirical quadratic Monge gap decreases to numerical zero. Right: the data error increases as monotonicity is imposed; the dashed line marks the hard monotone solution.

This confirms feasibility, but also exposes the modeling choice: the penalty does not merely smooth the regression curve. It favors maps whose sampled graph is optimal for quadratic transport, which on the line means increasing order.

4.3 What changes for a general convex h ?

The representer argument does not require a quadratic cost. For any empirical loss that depends on T only through $(T(x_i))_i$, a minimizer still has the form (21). With a convex displacement cost h , the finite objective becomes

$$\|Ka - y\|^2 + \lambda a^\top Ka + \frac{\gamma}{n} \sum_{i=1}^n [h(x_i - (Ka)_i) - h(x_i - (Ka)_{(i)})], \quad (25)$$

where $(Ka)_{(i)}$ denotes the i th order statistic. This is explicit and cheap to evaluate, but it is generally nonconvex. It is convex for every data set precisely when h is quadratic up to affine terms, under the smoothness assumptions of Corollary 2.2. More general affine-difference costs, such as reverse Bregman divergences, also retain a convex support-function formulation, but no longer reduce to the simple one-dimensional inversion hinge (17).

5 Conclusions

The main conclusions are structural rather than algorithm-specific.

1. The Monge gap is always nonnegative, but it is not convex for a general cost.
2. For smooth costs, universal convexity in the map is equivalent to affine dependence of every cross-cost difference on the image variable.
3. Within smooth displacement costs $h(x - y)$, this singles out quadratic costs, up to affine terms.
4. In one dimension, every convex h admits a quantile formula and gives a penalty for nonmonotonicity, but that penalty is generally nonconvex.
5. The one-dimensional quadratic gap is a rearrangement deficit and a permutahedral support function. Its proximal geometry is linked to isotonic optimization, and its RKHS-regularized regression problem is a finite convex program.

Thus the proposed kernel experiment is mathematically clean for the quadratic gap. For a nonquadratic convex displacement cost, the kernel trick still reduces the problem to finitely many coefficients, but it does not rescue convexity.

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